PUBLICATIONS

Peer-Reviewed

A novel electrocardiogram-synchronized laser Doppler holography system for cardiac cycle-resolved retinal hemodynamics: development and validation

Type: Peer-Reviewed Article (Published)
Author(s): Wood K, Schnorbus N, Muncharaz L, Lahoti Y, Siesky BA, Guidoboni G, Vercellin AV, Potash S, Greenberg LA, Atlan M, Chui TYP, Rosen R, Harris A
First/Senior Author: no
Publication Name: Scientific Reports
Issue Number: 1
Year: 2026
PubMed ID: 42071024
Publication Volume: 16
Pages: 20436
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/42071024/

Automated scoliosis X-ray cobb angle classification: a deep learning approach with RadImageNet

Type: Peer-Reviewed Article (Published)
Author(s): Yu J, Lahoti Y, Ahmed H, Kurapatti M, Frost J, Song J, Corvi J, Namiri N, Issa T, Cho S, Kim J
First/Senior Author: no
Publication Name: European Spine Journal
Issue Number: N/A
Year: 2026
PubMed ID: 42423737
Publication Volume: N/A
Pages: Epub ahead of print
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/42423737/

Normal Global Sagittal Alignment Radiographic Parameters in Patients Without Spinal Deformity

Type: Peer-Reviewed Article (Published)
Author(s): Duey AH, Hoang T, Lahoti YS, Kurapatti M, Yu A, Mohamed KS, Hoang R, Song J, Isleem UN, Li SQ, Kim JS, Cho SK
First/Senior Author: no
Publication Name: Global Spine Journal
Issue Number: 4
Year: 2026
PubMed ID: 41043809
Publication Volume: 16
Pages: 1736-1744
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/41043809/

Pelvic incidence correlates more strongly with proximal than distal lordosis and most strongly with L1-pelvic angle

Type: Peer-Reviewed Article (Published)
Author(s): Kurapatti M, Hoang R, Li A, Lahoti Y, Mohamed K, Yu A, Grewal A, Hoang T, Duey A, Lee J, Cho SK
First/Senior Author: no
Publication Name: Spine Deformity
Issue Number: 2
Year: 2026
PubMed ID: 41264220
Publication Volume: 14
Pages: 589-598
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/41264220/

Peripapillary Vessel Density Mediates the Relationship Between Axial Length and Visual Field Damage in Glaucoma

Type: Peer-Reviewed Article (Published)
Author(s): Potash S, Harris A, Verticchio Vercellin A, Wood K, Cohen GY, Rosen A, Kwon M, Rothstein J, Lahoti Y, Tanga L, Carnevale C, Siesky BA
First/Senior Author: no
Publication Name: Journal of Clinical Medicine
Issue Number: 9
Year: 2026
PubMed ID: 42123286
Publication Volume: 15
Pages: 3555
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/42123286/

Development of a novel machine learning model to automate vertebral column segmentation utilizing biplanar full-body imaging

Type: Peer-Reviewed Article (Published)
Author(s): Lahoti Y, Sai S, Ahmed W, Rajjoub R, Li M, Zaidat B, Cho SK, Kim JS
First/Senior Author: yes
Publication Name: The Spine Journal
Issue Number: 12
Year: 2025
PubMed ID: 40324481
Publication Volume: 25
Pages: 2803-2810
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/40324481/

Artificially Intelligent Billing in Spine Surgery: An Analysis of a Large Language Model

Type: Peer-Reviewed Article (Published)
Author(s): Zaidat B, Lahoti YS, Yu A, Mohamed KS, Cho SK, Kim JS
First/Senior Author: no
Publication Name: Global Spine Journal
Issue Number: 2
Year: 2025
PubMed ID: 38147047
Publication Volume: 15
Pages: 1113-1120
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/38147047/

Automated Scoliosis Cobb Angle Classification in Biplanar Radiograph Imaging With Explainable Machine Learning Models

Type: Peer-Reviewed Article (Published)
Author(s): Yu J, Lahoti YS, McCandless KC, Namiri NK, Miyasaka MS, Ahmed H, Song J, Corvi JJ, Berman DC, Cho SK, Kim JS
First/Senior Author: no
Publication Name: Spine
Issue Number: 13
Year: 2025
PubMed ID: 40007466
Publication Volume: 50
Pages: E259-E267
Website (if indicated): https://pubmed.ncbi.nlm.nih.gov/40007466/

An Electronic Health Record-Based Predictive Model to Quantify Multifactorial Risk for Primary Open Angle Glaucoma

Type: Peer-Reviewed Abstract (Published)
Author(s): Lahoti Y, Harris A, Pasquale LR, Sun M, Siesky BA, Cohen G, Wood Shalem K, Kwon M, Rosen A, Verticchio Vercellin A, Potash S, Wierzbowska J, Riina N, Sommi A, Belamkar AV, Guidoboni G
First/Senior Author: yes
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 3031
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2812417

Association Between Hemoglobin A1c and Vertical Cup-To-Disc Ratio in Adults Without Diabetes Mellitus

Type: Peer-Reviewed Abstract (Published)
Author(s): Liu S, Jaipalli S, Kang JH, Lahoti Y, Harris A, Pasquale LR
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 5134
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2817599

Artificial Intelligence Ready and Exploratory Atlas for Diabetes Insights: the association between glaucoma, hypertension, and cognitive performance

Type: Peer-Reviewed Abstract (Published)
Author(s): Greenfield J, Harris A, Lahoti Y, Gupta A, Pasquale LR, Siesky BA, Wood Shalem K, Cohen G, Wirostko BM, Potash S, Verticchio Vercellin A, Riina N, Harris D, Chang M, Safir F, Guidoboni G
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 2680
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2813198

Automated capillary-level optical coherence tomography angiography phenotyping in patients with glaucoma and diabetes

Type: Peer-Reviewed Abstract (Published)
Author(s): Lahoti Y, Guidoboni G, Greenfield J, Cohen GJ, Potash S, Siesky B, Verticchio Vercellin A, Wood K, Kwon M, Harris A
First/Senior Author: yes
Publication Name: Artificial Intelligence in Vision and Ophthalmology
Issue Number: 1 Suppl
Year: 2026
PubMed ID: N/A
Publication Volume: 2
Pages: 30
Website (if indicated): https://www.aivojournal.com/index.php/AIVO/article/view/166/270

Automated Deep-Learning FAZ Quantification and Cardiovascular Risk Associations in the AI-READI Cohort

Type: Peer-Reviewed Abstract (Published)
Author(s): Cohen G, Lahoti YS, Potash S, Gupta A, Pasquale LR, Wierzbowska J, Verticchio Vercellin A, Siesky BA, Ciulla TA, Tai TYT, Rothstein J, Isserow LJ, Harris A
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 2704
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2812438

Body fat composition and major outcomes in patients with perianal fistulizing Crohn's disease

Type: Peer-Reviewed Abstract (Published)
Author(s): Bergenholtz S, Cho WJ, Peraza J, Freid H, Lahoti Y, Blankemeier L, Fettahoglu A, Chaudhari A, Boutin R, Gold SL, Wong SY
First/Senior Author: no
Publication Name: Gastroenterology
Issue Number: 6
Year: 2026
PubMed ID: N/A
Publication Volume: 170
Pages: S-1009
Website (if indicated): https://doi.org/10.1016/S0016-5085(26)02623-5

Electrocardiogram-Synchronized Doppler Holography System for Quantifying Cardiac Cycle-Resolved Retinal Blood Flow

Type: Peer-Reviewed Abstract (Published)
Author(s): Wood Shalem K, Harris A, Muncharaz Duran L, Schnorbus N, Gurevitz C, Niv Granot Y, Buckley S, Lerakis S, Croft L, Lahoti YS, Sidoti PA, Siesky BA, Verticchio Vercellin A, Greenberg LA, Kanwar K, Riina N, Rosen RB
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 2356
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2812463

Ocular perfusion metrics demonstrate strong coupling with myocardial strain and pulmonary pressure

Type: Peer-Reviewed Abstract (Published)
Author(s): Siesky BA, Harris A, Potash S, Lahoti YS, Schanzer N, Niv Granot Y, Buckley S, Lerakis S, Croft L, Wood Shalem K, Verticchio Vercellin A, Oddone F, Carnevale C, Cohen G, Sacco R, Gurevitz C
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 2355
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2812464

Repeatability of retinal blood flow metrics measured using Doppler holography

Type: Peer-Reviewed Abstract (Published)
Author(s): Muncharaz Duran L, Greenberg LA, Najac MJ, Haq A, Schnorbus N, Wood Shalem K, Siesky BA, Verticchio Vercellin A, Lahoti Y, Auray Z, Fischer Y, Atlan M, Harris A, Rosen RB
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: N/A
Publication Volume: 67
Pages: 2904
Website (if indicated): https://arvojournals.org/article.aspx?articleid=2813185

Generative continuous-time recurrent neural networks for using irregularly sampled blood pressure dynamics to predict structural glaucoma progression

Type: Peer-Reviewed Abstract (Published)
Author(s): Riina N, Harris A, Whitley D, Siesky BA, Lahoti YS, Potash S, Verticchio Vercellin A, Cohen G, Wood Shalem K, Schanzer N, Arciero J, Fry BC, Eckert GJ, Vinod K, Januleviciene I, Guidoboni G
First/Senior Author: no
Publication Name: Investigative Ophthalmology & Visual Science
Issue Number: 7
Year: 2026
PubMed ID: NA
Publication Volume: 67
Pages: 584
Website (if indicated): https://iovs.arvojournals.org/article.aspx?articleid=2812310

Comprehensive comparison of sagittal alignment parameters in adult patients with and without scoliosis using full-body X-ray imaging

Type: Peer-Reviewed Abstract (Published)
Author(s): Kurapatti M, Hoang R, Grewal A, Lahoti Y, Yu A, Mohamed K, Duey A, Hoang T, Song J, Lee J, Cho SK
First/Senior Author: no
Publication Name: The Spine Journal
Issue Number: 11 Suppl
Year: 2025
PubMed ID: N/A
Publication Volume: 25
Pages: S108
Website (if indicated): https://www.thespinejournalonline.com/article/S1529-9430(25)00453-X/abstract

Automated calculation and estimation of global sagittal alignment from weight-bearing full-body X-ray imaging using deep learning

Type: Peer-Reviewed Abstract (Published)
Author(s): Kurapatti M, Lahoti Y, Guttikonda S, Etigunta S, Yu A, Mohamed K, Ahmed W, Cho SK
First/Senior Author: no
Publication Name: The Spine Journal
Issue Number: 11 Suppl
Year: 2025
PubMed ID: N/A
Publication Volume: 25
Pages: S72
Website (if indicated): https://www.thespinejournalonline.com/article/S1529-9430(25)00725-9/abstract

Reference global sagittal alignment parameters stratified by age and sex in patients without spinal deformity

Type: Peer-Reviewed Abstract (Published)
Author(s): Kurapatti M, Duey A, Hoang R, Lahoti Y, Mohamed K, Yu A, Hoang T, Song J, Cho SK
First/Senior Author: no
Publication Name: The Spine Journal
Issue Number: 11 Suppl
Year: 2025
PubMed ID: N/A
Publication Volume: 25
Pages: S123-S124
Website (if indicated): https://www.thespinejournalonline.com/article/S1529-9430(25)00486-3/abstract

Use of an automated, deep learning software to evaluate body composition including skeletal muscle mass and visceral adiposity in patients with inflammatory bowel disease

Type: Peer-Reviewed Abstract (Published)
Author(s): Gold SL, Blankemeier L, Lahoti Y, Chaudhari A, Sands BE, Boutin R
First/Senior Author: no
Publication Name: Gastroenterology
Issue Number: 5 Suppl
Year: 2024
PubMed ID: N/A
Publication Volume: 166
Pages: S-1484-S-1485
Website (if indicated): https://doi.org/10.1016/S0016-5085(24)03853-8

Automated scoliosis classification from EOS full body imaging through artificial intelligence: a deep learning approach with RadImageNet

Type: Peer-Reviewed Abstract (Published)
Author(s): Yu J, Lahoti Y, Cho SK, Kim JS
First/Senior Author: no
Publication Name: The Spine Journal
Issue Number: 9 Suppl
Year: 2024
PubMed ID: N/A
Publication Volume: 24
Pages: S81
Website (if indicated): https://www.thespinejournalonline.com/article/S1529-9430%2824%2900363-2/fulltext

A Scalable Framework for Harmonized mtDNA Analysis Across Diverse Biobanks

Type: Peer-Reviewed Article (Not Published)
Author(s): Schecter D, Lee S, Vimal T, Lahoti Y, Goncalves V, Retallick-Townsley K, Pang J, Guvenek A, Preuss M, Tinker R, Morava E, Kozicz T, Hirano M, Ganesh J, Naini A, Liang J, Davis L
First/Senior Author: no
Publication Name: Nature Methods
Publication Status: Submitted

Artificial Intelligence Accurately Assists in Billing for Orthopaedic Lower Extremity Surgery: Performance of the Mistral-NeMo Language Model

Type: Peer-Reviewed Article (Not Published)
Author(s): Li M, Guttikonda S, Lahoti Y, Sheetz M, Isleem U, Cho SK, Kim JS
First/Senior Author: no
Publication Name: Arthroscopy
Publication Status: In Press

Central Retinal Artery Hemodynamics as a Non-Invasive Marker of Cardiac Diastolic Function

Type: Peer-Reviewed Article (Not Published)
Author(s): Harris A, Wood K, Verticchio Vercellin A, Niv-Granot Y, Lerakis S, Croft L, Gurevitz C, Siesky BA, Lahoti Y, Riina N, Olayinka T, Guidoboni G
First/Senior Author: no
Publication Name: Translational Vision Science & Technology
Publication Status: Submitted

Association between glaucoma and cognitive performance in a large-scale multimodal ophthalmic cohort from the AI-READI consortium

Type: Peer-Reviewed Article (Not Published)
Author(s): Lahoti Y, Harris A, Greenfield J, Riina N, Siesky B, Pasquale L, Cohen G, Gupta A, Minkis J, Kwon M, Wood K, Cantor B, Elahi F, Guidoboni G
First/Senior Author: yes
Publication Name: American Journal of Ophthalmology
Publication Status: Submitted

Semi-automated measurement of central retinal artery blood flow using Doppler ultrasound coupled with electrocardiography

Type: Peer-Reviewed Article (Not Published)
Author(s): Harris A, Wood K, Olayinka T, Verticchio Vercellin A, Niv-Granot Y, Lerakis S, Gurevitz C, Siesky BA, Lahoti Y, Guidoboni G
First/Senior Author: no
Publication Name: Translational Vision Science & Technology
Publication Status: In Press

Transfer learning methods from clinical research to population-based studies in glaucoma

Type: Peer-Reviewed Article (Not Published)
Author(s): Joyce M, Rai R, Guidoboni G, Keller J, Wikle C, Lahoti Y, Chong RS, Cheng CY, Siesky B, Cohen GY, Harris A
First/Senior Author: no
Publication Name: British Journal of Ophthalmology
Publication Status: In revision

Presentations

Using dynamic time warping to find patterns in coronal spinal alignment

Type: Oral Presentation
Author(s)/Presenter(s): Lahoti Y, Yu J, Cho SK, Kim JS
First/Senior Author: yes
Event/Meeting: Eastern Orthopaedic Association Annual Meeting
Address: Palm Beach, Florida, United States
Month/Year: 10/2026

Integrating computational modeling and artificial intelligence for improved diagnosis and longitudinal follow-up in ophthalmology

Type: Oral Presentation
Author(s)/Presenter(s): Lahoti Y., Harris A
First/Senior Author: yes
Event/Meeting: SIAM Conference on the Life Sciences
Address: Cleveland, Ohio, United States
Month/Year: 07/2026

Global sagittal alignment variations with body mass index in patients

Type: Oral Presentation
Author(s)/Presenter(s): Mohamed K, Duey AH, Yu A, Kurapatti MA, Hoang R, Lahoti Y, Hoang T, Song J, Li SQ, Cho SK
First/Senior Author: no
Event/Meeting: American Academy of Orthopaedic Surgeons Annual Meeting
Address: Orlando, Florida, United States
Month/Year: 10/2025

Automated scoliosis classification from AI-enabled spine contouring and Cobb angle estimation

Type: Oral Presentation
Author(s)/Presenter(s): Lahoti Y, Yu J, Cho SK, Kim JS
First/Senior Author: yes
Event/Meeting: Eastern Orthopaedic Association Annual Meeting
Address: Palm Beach, Florida, United States
Month/Year: 10/2024